

HACKATHON PROJECT SUBMISSION

FarmFleet

Localized Agricultural Equipment Sharing & Marketplace
Platform

PROJECT REPORT

SUBMITTED BY

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Abstract

Agriculture plays a significant role in India's economy, yet many small and marginal farmers cannot afford modern agricultural machinery because of high purchase costs. As a result, they often rely on manual labor or outdated equipment, which reduces productivity and increases operational expenses. At the same time, owners of agricultural machinery frequently have idle equipment that remains unused for long periods.

FarmFleet is a localized agricultural equipment sharing platform designed to bridge this gap by connecting equipment owners with farmers who need machinery on a temporary basis.

The platform enables users to search for nearby agricultural equipment, compare rental prices, communicate directly with equipment owners, and make booking requests.

The system is developed using a PHP REST API backend, SQLite database, HTML5, CSS3, and Vanilla JavaScript. The application supports multilingual functionality, responsive design, GPS-based equipment search, and WhatsApp integration for communication.

FarmFleet reduces equipment costs, improves machinery utilization, and promotes digital transformation in rural agriculture.

Introduction

Mechanization is one of the major factors influencing agricultural productivity. However, purchasing tractors, harvesters, sprayers, and other agricultural machinery is financially impossible for many small farmers.

Traditional equipment rental methods rely on local brokers, personal contacts, and manual negotiations, making the process inefficient and expensive.

FarmFleet introduces a digital marketplace where farmers can quickly discover nearby agricultural equipment available for rent. Equipment owners can list their machinery with pricing details, availability, and location information. Farmers can filter equipment based on category, distance, and rental cost before contacting the owner directly.

The platform is designed specifically for rural users by providing a simple interface, multilingual support, high-contrast visuals, and mobile-first responsiveness, reflecting the goals outlined in the SRS.

Problem Statement and Objectives

PROBLEM STATEMENT

Small and marginal farmers experience several challenges:

- High machinery purchase cost
- Limited access to nearby equipment
- Dependence on middlemen
- Lack of pricing transparency
- No real-time availability information
- Poor communication between owners and farmers

Equipment owners also face low machinery utilization because their equipment remains unused for much of the year.

OBJECTIVES

The objectives of FarmFleet are:

- Develop an online agricultural equipment rental platform.
- Enable location-based equipment discovery.
- Support hourly and daily rental options.
- Provide bilingual support (English and Tamil).
- Simplify communication through WhatsApp.
- Increase equipment utilization.
- Eliminate unnecessary broker commissions.
- Offer a secure and responsive web application.

System Comparison

Existing System

The traditional agricultural equipment rental process has several limitations:

- Manual booking
- Phone-call based communication
- No centralized equipment listings
- Difficult price comparison
- No availability tracking
- Heavy dependence on local contacts

Disadvantages

- Time consuming
- Expensive
- Limited equipment choices
- Poor transparency
- High commission charges

Proposed System

FarmFleet introduces a digital marketplace that provides:

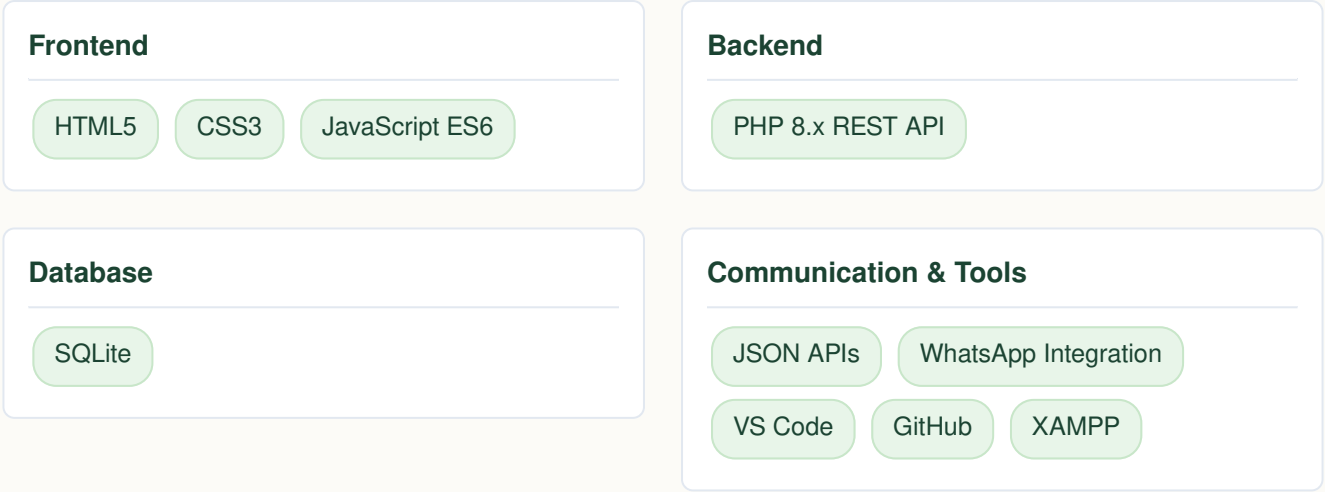
- Online equipment listings
- GPS-based nearby search
- Equipment categorization
- Real-time availability
- Owner contact through WhatsApp
- Mobile-friendly interface
- Secure user authentication

Advantages

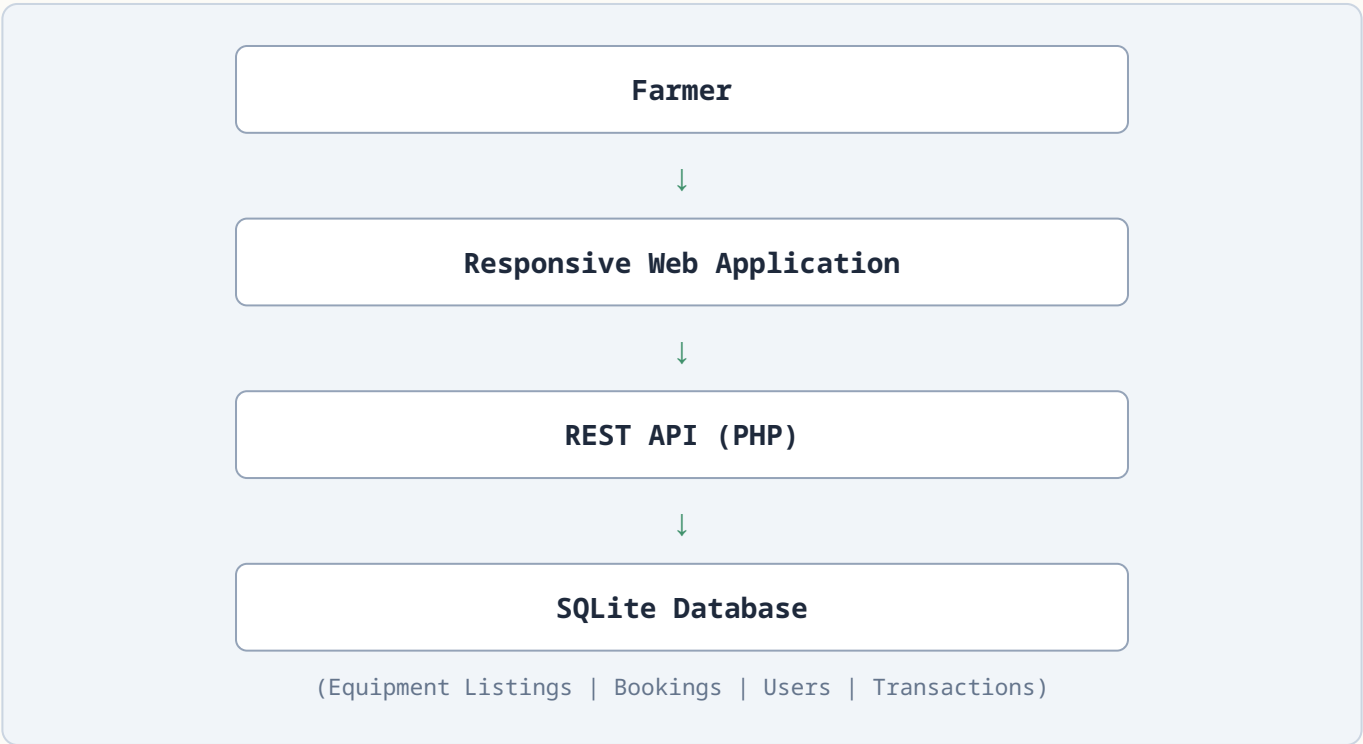
- Easy booking process
- Transparent pricing
- Better equipment utilization
- Reduced operational cost
- Improved accessibility

Technology Stack & System Architecture

TECHNOLOGY STACK



SYSTEM ARCHITECTURE



The client sends requests to the REST API. The API processes the request and communicates with the SQLite database before returning JSON responses to the frontend.

Database Design and Modules

DATABASE TABLES

Table Name	Fields
Users	User ID, Name, Phone Number, Role, Password
Equipment	Equipment ID, Name, Category, Rental Price, Location, Availability
Bookings	Booking ID, Farmer ID, Equipment ID, Duration, Status
Transactions	Transaction ID, Booking ID, Amount, Commission

SYSTEM MODULES

Farmer Module

- Register/Login
- Search Equipment
- Book Equipment
- View History

Owner Module

- Add Equipment
- Edit Listings
- Accept Requests
- View Earnings

Admin Module

- Verify Users
- Manage Listings
- Platform Statistics
- Commission Settings

Features and Security

KEY FEATURES

- Equipment Search
- Dynamic Filtering
- GPS Location
- WhatsApp Contact
- Responsive Design
- Tamil Language Support
- Booking History
- Owner Dashboard
- Admin Dashboard

SECURITY FEATURES

- Role-Based Access Control
- Session Authentication
- SQL Injection Prevention
- Prepared Statements
- Input Validation
- Secure Password Storage
- Phone Number Verification

Testing, Results & Enhancements

TESTING

- **Unit Testing:** Individual modules were tested independently.
- **Integration Testing:** API communication between frontend and backend was verified.
- **Security Testing:** Session validation, login authentication, and SQL injection prevention were verified.
- **UI Testing:** Tested on mobile devices, desktop browsers, and tablet compatibility.

RESULTS

The developed system successfully:

- Displays equipment listings
- Supports booking requests
- Performs location-based searches
- Maintains user sessions
- Provides multilingual support
- Calculates commissions correctly

FUTURE ENHANCEMENTS

Android Application

iOS Application

Online Payments

AI Equipment Recommendation

IoT Integration

PostgreSQL Migration

Cloud Deployment

Conclusion and References

CONCLUSION

FarmFleet is a modern agricultural equipment sharing platform designed to improve accessibility to farm machinery for small and marginal farmers. By providing location-based equipment discovery, responsive web access, secure authentication, and direct owner communication, the system helps reduce rental costs while increasing equipment utilization.

The project demonstrates how web technologies such as PHP, SQLite, and JavaScript can be used to create an efficient, scalable, and user-friendly agricultural marketplace. With future enhancements such as mobile applications, online payments, and IoT integration, FarmFleet has the potential to support wider adoption of digital agriculture.

End of Project Report — FarmFleet 2026